ACCESS KNOB: PARAMETRIC DESIGN OF 3D-PRINTABLE ACCESSIBLE SYNTHESIZER CONTROLS

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ASSETS '26
The 28th International ACM SIGACCESS Conference
Athens, Greece • October 26-28, 2026

INTRODUCTION & CO-DESIGN CONTEXT

Commercial synthesizer interfaces present significant accessibility barriers for musicians with motor and sensory impairments. Standard controls, which are typically small, smooth, and closely spaced, require a high-precision pinch grip and considerable physical force to actuate. Over time, millions of musicians experience hand function and grip strength decline due to aging, arthritis, stroke, or neuropathies.

> *"Dialysis-dependent kidney failure caused severe hand neuropathy in a retired IT worker and synth collector, leaving his collection of 20 hardware synthesizers completely unplayable. Traditional workarounds like using a T-wrench or palm-dragging were fatiguing and lacked expressive nuance."*

This lived co-design encounter inspired **ACCESS KNOB**, an open-source browser-based parametric customizer. Built on Universal Music Design (UMD) principles, the project shifts the paradigm from mass-produced controls to personalized, user-fabricated interfaces that enhance musical expression, independence, and creative control.

PROCEDURAL WASM CAD KERNEL

Traditional CAD tools present steep learning curves and lack accessibility. ACCESS KNOB utilizes a client-side WebAssembly-compiled **Manifold** geometry kernel to perform boolean operations in milliseconds, enabling real-time parameter tweaking directly in the browser.

Two Mounting Modes & Three Spindle Bores:

- **Swap-In Mode:** Replaces the original manufacturer knob entirely. Generates custom internal flat/splined bores to slide flush onto the bare metal/plastic spindle.
- **Slide-Over Mode:** An adaptive sleeve that slides snugly over the original control cap, preserving the instrument's resale value and avoiding complex hardware disassembly.
- **Spindle Bore Profiles:** Generates custom flats/splines for **D-Shafts**, **Knurled/Splined** (18/24-tooth gears), and **Solid Round** spindles (secured via integrated M3 set-screw).

Figure 4: Small, high-density configurations showing how varied diameters and shape options fit compact Eurorack or desktop synthesizer panels, utilizing M3 set-screws for secure shaft locking.

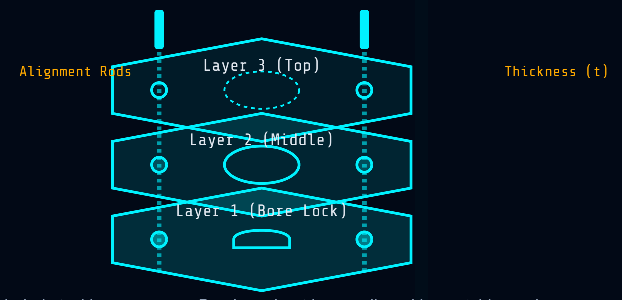
2D VECTOR SVG STACKING EXPORTER

To democratize fabrication for makerspaces without access to 3D printers, ACCESS KNOB includes a 2D vector SVG stacking exporter. This tool enables subtractive manufacturing (laser-cutting or CNC routing) using sheet materials (wood, acrylic, bamboo, or composites).

> *"The exporter dynamically slices the 3D model into layer profiles, compensates for laser kerf, and adds kerf-compensated slot paths and stacking rods."*

System Assembly Mechanics:

- **Kerf Compensation:** Sub-millimeter adjustment (default 0.1 mm) ensures slots and outer boundaries match target shapes after laser-cutting.
- **Dual Alignment Rods:** Generates two rods of width `thickness` and length `heightKnob + kerf`. Slits are cut through all stacked layers, allowing the rods to slide through, aligning the pieces perfectly for gluing.
- **Layer Taper:** Outer contours are calculated iteratively at height intervals to preserve the overall tapered ergonomic profile.



Exploded stacking sequence: Bamboo sheet layers aligned by matching rods

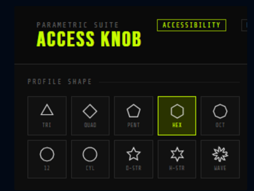


Figure 1: Procedural geometric profile options (triangle, square, hexagon, octagon, star, and gear) for tactile shape-coding, aiding blind or low-vision users in distinguishing knob functions by touch.

TACTILE MATERIAL DYNAMICS

Material choice impacts physical sensation, thermal stability, and control responsiveness during musical play:

Material	Process	Rigidity	Friction	Tactile Feedback & Damping
PLA	FDM 3D Print	High	Medium	Crisp edges; low vibrational damping
PETG	FDM 3D Print	Medium	Medium	Slight elasticity (ideal for outer sleeves)
TPU	FDM 3D Print	Very Low	High	Flexible; excellent non-slip inner lining
Bamboo	Laser-Cut Stack	High	High (Grain)	Organic feel; superior mechanical damping

MECHANICAL EVALUATION

To evaluate how parametric adjustments reduce required physical force, we model the peripheral finger force F required to rotate a potentiometer with torque resistance T:

$$T = F \cdot (D / 2) \Rightarrow F = 2T / D$$

Increasing the outer diameter D from a standard 10 mm to an accessible 32 mm or 40 mm reduces the required finger pinch force by **68.8% to 75.0%**, compensating directly for reduced grip strength.

Diameter (D)	Required Force (F)	Force Reduction
10 mm (Standard)	$F_{10} = 2T / 10$	0.0% (Baseline)
15 mm	$0.67 \cdot F_{10}$	33.3%
25 mm (Medium)	$0.40 \cdot F_{10}$	60.0%
32 mm (Large)	$0.31 \cdot F_{10}$	68.8%
40 mm (Accessible)	$0.25 \cdot F_{10}$	75.0%

Grip Shear Force: For musicians utilizing palm-dragging or side-of-hand shear techniques, the functional friction force F_f driving rotation is modeled as:

$$F_f = \mu \cdot F_n \quad \cdot \quad A = \pi \cdot D \cdot H$$

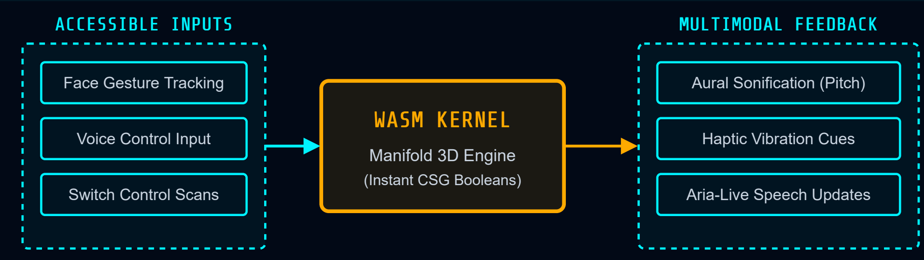
where μ is the material's friction coefficient, and F_n is the normal force. Taller knobs expand contact area A, distributing downward pressure to enable actuation with minimal pinch grasp requirements.

Figure 3: Star-pattern flutes and deep grooves on custom ACCESS KNOBs. The pronounced edges allow palm-dragging and side-of-hand shear techniques, dramatically increasing rotational torque without requiring a tight pinch.

ACCESSIBILITY & FUTURE WORK

Hands-Free Design Customization:

The web customizer features high-contrast glowing visual focus states, supporting commodity webcam head-tracking and facial gesture interfaces (Google Project Gameface, Apple Voice/Switch Control).



Future Work & Clinical Evaluation:

We are expanding the customizer to generate fader caps and drum pads. We propose an IRB-approved protocol with $N = 12$ musicians with hand impairments to assess custom slide-over vs. swap-in controls. Evaluation metrics will include task completion times, error rates (such as accidental adjacent knob nudging), and System Usability Scale (SUS) scores to evaluate creative agency.

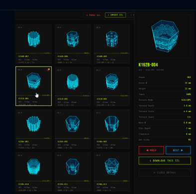


Figure 2: Procedural customizer interface showing a batch of 12 generated variants, the real-time WebGL view, and the parameter sliders.